

IMEN266 Operations Research II — Homework 2 (Ch. 4, Markov chains)

Fall 2026

Ungraded, but: **some Part-A problems may reappear on the exams.**

Submit to PLMS: one PDF (parts A, B, D write-ups) + your HW2.ipynb.

AI rules for this homework (details: docs/ai_usage_guide.pdf)

- **Part A:** solve *alone first*, closed-book style, on paper. Only afterwards may you check with an AI — and if the AI changes your mind, record why in Part D.
- **Parts B–C:** AI collaboration is expected. You remain responsible for every claim you submit.
- **Part D:** the prompt log is mandatory whenever AI was used.

Part A — Exam-style warm-up (no AI on first attempt)

- A1. (**Modeling.**) A production cell has two identical machines and *one* repair crew. Each working machine fails during a day with probability 0.1, independently of everything else. A broken machine that the crew works on is repaired by the next morning with probability 0.6; the crew can work on only *one* machine per day. Let X_n be the number of working machines at the start of day n . Model $\{X_n, n \geq 0\}$ as a DTMC: state space, transition matrix (with the reasoning behind each row), and transition diagram.
- A2. (**Transient analysis.**) A DTMC on $\{1, 2, 3\}$ has

$$P = \begin{pmatrix} 0.2 & 0.5 & 0.3 \\ 0.4 & 0.4 & 0.2 \\ 0.1 & 0.3 & 0.6 \end{pmatrix}, \quad a = (0.5, 0.5, 0).$$

Compute (i) $P(X_2 = 3)$; (ii) $P(X_0 = 1, X_1 = 2, X_2 = 2)$; (iii) $P(X_4 = 1 \mid X_2 = 3)$; (iv) $E[X_2]$. Show the matrix products you use.

- A3. (**Limiting behavior, cost.**) For the chain of A1: (i) Is it irreducible? aperiodic? ergodic? Justify briefly. (ii) Find the stationary distribution π (a 3×3 system; give exact fractions or 4 decimals). (iii) In the long run, what fraction of days start with both machines down, and what is the mean number of days between two such days? (iv) A working machine earns 100 (thousand won) per day; the crew costs 30 per day whenever at least one machine is broken. Compute the long-run average daily profit.
- A4. (**Classification.**) States $1, \dots, 5$,

$$P = \begin{pmatrix} 0.5 & 0.5 & 0 & 0 & 0 \\ 0.5 & 0.5 & 0 & 0 & 0 \\ 0.2 & 0 & 0.3 & 0.5 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 \end{pmatrix}.$$

Give the communicating classes, say which are recurrent/transient, the period of each class, and whether $\lim_{n \rightarrow \infty} P^n$ exists. Describe *all* stationary distributions.

Part B — Critique an AI solution (find the flaws)

A student pasted the Himart problem (CP3 #3: Monday stock, order-up-to 5 when Friday stock < 2 , weekly demand $\sim \text{Poisson}(3)$, lost sales) into an AI assistant and asked:

“What is the long-run probability that an order is placed at the end of a week? And if the store starts with 5 PCs, what is the probability that there are exactly 3 PCs at the beginning of the 4th week?”

The AI replied:

- Step 1. Let the state be the Monday stock, $S = \{2, 3, 4, 5\}$. From state i the next state is $i-d$ with probability $P(D=d)$ for $d=0, \dots, i-2$, and it is 5 with probability $P(D=i-1)$ (the demand that brings the stock below 2).
- Step 2. The rows of this matrix do not sum to one because of rounding in the Poisson table, so I normalise each row to sum to 1.
- Step 3. The stationary distribution solves $P\pi = \pi$; the eigenvector of P for eigenvalue 1 is proportional to $(1, 1, 1, 1)$, so $\pi = (0.25, 0.25, 0.25, 0.25)$.
- Step 4. An order brings the stock to 5, so the long-run probability of an order is $\pi_5 = 0.25$.
- Step 5. After 4 weeks the chain has reached steady state, so $P(X_4 = 3) = \pi_3 = 0.25$.

Your tasks:

- B1. Identify **at least three distinct errors** in this reasoning. Name each precisely (which probability is confused with which; which equation is transposed; which event is mis-specified; which limit is misused). *One of the errors is a modeling error that a “fix” then hides — explain why normalising was the wrong reaction.*
- B2. Write the correct solution: the transition matrix with exact Poisson expressions (tail probabilities where needed), the stationary distribution (numerical, from a linear system), the long-run probability of an order as $\sum_i \pi_i P(\text{order} \mid i)$, and $P(X_4 = 3 \mid X_0 = 5)$ from aP^4 . You may use `HW2.ipynb` for the numbers, but the formulas must be on paper.
- B3. Verify both numbers by simulating the *store policy* directly (demand $\rightarrow$ Friday stock $\rightarrow$ order decision; a skeleton cell is provided). Report the estimates and 95 % confidence intervals.

Part C — Stress-test the assumption (AI + simulation)

CP3 #2 assumes tomorrow’s weather depends only on today’s. Suppose the truth is the *spell* model of CP3 #7 with two types (sunny/rainy), $p_k = (0.6, 0.7, 0.8, 0.9)$ and $q_k = (0.5, 0.4, 0.3, 0.2)$ for the k -th day of a spell (persistence constant from day 4 on), but you only observe the daily sequence of S/R and, as a modeler, you **fit a 2-state DTMC** by transition frequencies: $\hat{P}_{ij} = \#\{i \rightarrow j\} / \#\{i\}$.

- C1. **Before simulating**, discuss with your AI assistant which of the following the fitted chain will reproduce correctly, and write a one-line reason for each prediction: (a) the long-run fraction of sunny days; (b) the *mean* length of a sunny spell; (c) $P(\text{sunny in 3 days} \mid \text{sunny today})$; (d) the *distribution* of sunny-spell lengths; (e) $P(\text{sunny tomorrow} \mid \text{sunny today and yesterday})$.
- C2. Simulate 100,000 days from the spell model (skeleton provided), fit $\hat{P}$, and compare (a)–(e) between truth and fit. Explain the pattern of hits and misses in at most five sentences. *Moral: a Markov fit matches one-step statistics exactly; every multi-step statement is a modeling assumption.*

- C3. (Optional) The PageRank practicum used $\beta = 0.85$. For the 200-page random web, plot how the top-10 set changes as β runs from 0.5 to 0.99 (e.g. overlap with the $\beta = 0.85$ top-10). Where does the ranking become unstable, and what happens as $\beta \rightarrow 1$?

Part D — Reflection and prompt log (mandatory if AI was used)

Fill the table in `HW2.ipynb` (or reproduce it in your PDF): one row per meaningful AI interaction — where you used it, the prompt, what it returned, and **what you checked or corrected**. Then answer in 2–3 sentences: *where was the AI most useful, and where was it least trustworthy, on this homework?*

Logistics. Solutions to Part A will be posted on PLMS about 10 days after release. Parts B–D have no single “answer key”; selected strong submissions (anonymized) will be discussed in class.